



THESIS ASSIGNMENT

Name and Surname: Bc. Tomáš Barančok
Study programme: Applied Computer Science (Single degree study, master II. deg., full time form)
Field of Study: Computer Science
Type of Thesis: Diploma Thesis
Language of Thesis: English
Secondary language: Slovak

Title: Simulation and Synthetic Data Generation of Deformable Objects

Annotation: This research focuses on generation of synthetic 3D data of deformable objects. An enormous amount of 2D annotated datasets is available over the internet, however 3D datasets focus on a specific 3D capturing hardware are limited. Moreover, in the case of deformable objects, the annotation process itself is hard to perform on real 3D scans and also they are hard to simulate and render. An example of deformable objects is packing of clothing in a bin-picking application or plastic waste in a waste-sorting application.

Keywords: Synthetic data generation, deformable objects, transparent objects, clothing packing, waste sorting, simulation, digital twin

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Department: FMFI.KAI - Department of Applied Informatics
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Approved: 13.10.2025 prof. RNDr. Roman Ďurikovič, PhD.
Guarantor of Study Programme

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